

Example 15.7

A company with 1 million shares worth \$40 each is considering issuing 200,000 warrants each giving the holder the right to buy one share with a strike price of \$60 in 5 years. It wants to know the cost of this. The interest rate is 3% per annum, and the volatility is 30% per annum. The company pays no dividends. From equation (15.20), the value of a 5-year European call option on the stock is \$7.04. In this case, $N = 1,000,000$ and $M = 200,000$, so that the value of each warrant is

$$\frac{1,000,000}{1,000,000 + 200,000} \times 7.04 = 5.87$$

or \$5.87. The total cost of the warrant issue is $200,000 \times 5.87 = \$1.17$ million. Assuming the market perceives no benefits from the warrant issue, we expect the stock price to decline by \$1.17 to \$38.83.

15.11 IMPLIED VOLATILITIES

The one parameter in the Black–Scholes–Merton pricing formulas that cannot be directly observed is the volatility of the stock price. In Section 15.4, we discussed how this can be estimated from a history of the stock price. In practice, traders usually work with what are known as *implied volatilities*. These are the volatilities implied by option prices observed in the market.¹⁰

To illustrate how implied volatilities are calculated, suppose that the market price of a European call option on a non-dividend-paying stock is 1.875 when $S_0 = 21$, $K = 20$, $r = 0.1$, and $T = 0.25$. The implied volatility is the value of σ that, when substituted into equation (15.20), gives $c = 1.875$. Unfortunately, it is not possible to invert equation (15.20) so that σ is expressed as a function of S_0 , K , r , T , and c . However, an iterative search procedure can be used to find the implied σ . For example, we can start by trying $\sigma = 0.20$. This gives a value of c equal to 1.76, which is too low. Because c is an increasing function of σ , a higher value of σ is required. We can next try a value of 0.30 for σ . This gives a value of c equal to 2.10, which is too high and means that σ must lie between 0.20 and 0.30. Next, a value of 0.25 can be tried for σ . This also proves to be too high, showing that σ lies between 0.20 and 0.25. Proceeding in this way, we can halve the range for σ at each iteration and the correct value of σ can be calculated to any required accuracy.¹¹ In this example, the implied volatility is 0.235, or 23.5%, per annum. A similar procedure can be used in conjunction with binomial trees to find implied volatilities for American options.

Implied volatilities are used to monitor the market's opinion about the volatility of a particular stock. Whereas historical volatilities (see Section 15.4) are backward looking, implied volatilities are forward looking. Traders often quote the implied volatility of an option rather than its price. This is convenient because the implied volatility tends to be less variable than the option price. The implied volatilities of actively traded options on an asset are often used by traders to estimate appropriate implied volatilities for other options on the asset.

¹⁰ Implied volatilities for European and American options can be calculated using DerivaGem.

¹¹ This method is presented for illustration. Other more powerful methods, such as the Newton–Raphson method, are often used in practice (see footnote 3 of Chapter 4).

The VIX Index

The CBOE publishes indices of implied volatility. The most popular index, the SPX VIX, is an index of the implied volatility of 30-day options on the S&P 500 calculated from a wide range of calls and puts. It is sometimes referred to as the “fear factor.” An index value of 15 indicates that the implied volatility of 30-day options on the S&P 500 is estimated as 15%. Information on the way the index is calculated is in Section 26.16. Trading in futures on the VIX started in 2004 and trading in options on the VIX started in 2006. One contract is on 1,000 times the index.

Example 15.8

Suppose that a trader buys an April futures contract on the VIX when the futures price is 18.5 (corresponding to a 30-day S&P 500 volatility of 18.5%) and closes out the contract when the futures price is 19.3 (corresponding to an S&P 500 volatility of 19.3%). The trader makes a gain of \$800.

A trade involving options on the S&P 500 is a bet on the future level of the S&P 500, which depends on the volatility of the S&P 500. By contrast, a futures or options contract on the VIX is a bet only on volatility. Figure 15.4 shows the VIX index between January 2004 and June 2020. Between 2004 and mid-2007 it tended to stay between 10 and 20. It reached 30 during the second half of 2007 and a record 80 in October and November 2008 after Lehman’s bankruptcy. By early 2010, it had declined to more normal levels. It has spiked several times since 2010 because of stresses and uncertainties in financial markets. In 2020, there was another big increase because of the COVID-19 pandemic.

VIX monitors the volatility of the S&P 500. The CBOE publishes a range of other volatility indices. These are on other stock indices, commodity indices, interest rates, currencies, and some individual stocks (for example, Amazon and Goldman Sachs). There is even a volatility index of the VIX index (VVIX).

Figure 15.4 The VIX index, January 2004 to June 2020.

